Multi-View Basketball Player and Ball Tracking with 3D Reconstruction

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Abstract. We present a multi-camera pipeline for tracking players, referees and the ball during a basketball action recorded at the Sanbapolis facility, and for reconstructing their 3D trajectories via triangulation. Ground truth is annotated at 5 fps on three synchronized views (2846 boxes, 309 frames, 16 categories). Players/referees are tracked per camera at the native 25 fps with a YOLO detector and its built-in tracker, reaching Precision = 0.775, Recall = 0.712 (F1 = 0.742), MOTP = 0.423 across all three views. The ball is far harder: a classical motion/colour baseline and a zero-shot pretrained small-ball detector (WASB/HRNet, with tiling on the widest-shot camera) reach F1 = 0.068 and F1 = 0.143 respectively – the latter roughly double, but both still limited by the ball's small size (a first fine-tuned-detector attempt failed for the same structural reason and was abandoned). Camera intrinsics/extrinsics were provided for this project; 2D tracks are re-identified across views by a reprojection-error voting scheme and triangulated into a court-centered frame. Against pseudo-ground truth from triangulated annotations, the reconstruction reaches a median error of 0.72 m (RMSE 1.01 m) over 44.9% of candidate points, and the 18 resulting 3D tracks keep a valid position in 88.1% of the frames within their own span on average.

1 Introduction

This work addresses the Player Tracking project of the Computer Vision course: given a multi-camera setup at the Sanbapolis facility, detect and track players, referees and the ball in each view, and reconstruct their 3D trajectories by triangulation. One action was recorded from three fixed, synchronized cameras (`out2`, `out4`, `out13`) on a FIBA-regulation 28×15 m court. As a team of two, the mandatory scope is annotation (task 1), 2D tracking with evaluation (task 2/2a), and 3D triangulation with evaluation (task 3/3a) using at least two views; we use all three throughout.

Two aspects make this work not straight-forward. The ball is small and fast, frequently occluded or blurred, and measures well under 20 px after the mandatory resize to a detector's input resolution – comparable to a CNN backbone's stride. And cross-camera association cannot reuse a per-camera ID: correspondence has to come from projective geometry, since no appearance-based re-identification was used (future work, Sec. 4).

This report covers the dataset, the camera model, the 2D detection/tracking pipeline (players and ball), and cross-camera 3D triangulation, each evaluated quantitatively − including a trajectory-level continuity measure − against the annotated ground truth (Sec. 3).

2 Method

2.1 Dataset and Annotation

One action was annotated from the three views on Roboflow at 5 fps: 309 frames (99/105/105 for `out2`/`out4`/ `out13`) and 2846 boxes over 16 categories – a generic `person` catch-all, `Ball` (191 boxes), six jersey-numbered "Red" identities and six "White" (two of the twelve, `Red_13`/`White_11`, appear in only 1/5 frames and are edge cases rather than consistently tracked identities), and two referees. Per-identity annotation lets the 3D pseudo-ground-truth (Sec. 3.3) match a specific person across cameras without a separate re-identification step; for the 2D evaluation (Sec. 3.2) every non-ball category is pooled into one `player` class.

2.2 Camera Model

Per-camera intrinsics (K, distortion) and extrinsics (rotation/ translation w.r.t. a court-centered frame) were provided for this project rather than self-calibrated. They undistort every frame before 2D detection (Sec. 2.3) and build each camera's projection matrix $P = K[R\,|\,t]$ for triangulation (Sec. 2.4); we do not report a separate reprojection-error figure for the calibration itself, since it was not computed by us – the full undistortion+ triangulation chain is instead validated end-to-end in Sec. 3.3.

2.3 2D Detection and Tracking

Players and referees. Each camera is processed independently at the native 25 fps: frames are undistorted (a per-pixel remap built once, reused every frame), resized to 1920×1088, and passed to a YOLO detector (COCO `person`/`sports ball` classes [6]) run through Ultralytics' tracking mode (`persist=True`), which defaults to BoT-SORT [3] when no alternative tracker is configured, as here. A low confidence threshold (0.1) favors recall, relying on the tracker's temporal persistence to filter spurious detections.

Ball. The same pipeline is largely blind to the ball: at 1920×1088 it covers a handful of pixels, below what the backbone preserves after downsampling. A first attempt fine-tuned YOLO on ball-only crops with SAHI tiled inference [2]; this did not fix recall, which is structural (the ball shrinks to single-digit pixel counts even per tile) rather than a training issue. Two alternatives were evaluated instead: (a) a classical baseline (MOG2 background subtraction with shadow removal, morphological cleanup, a circularity/aspect-ratio shape filter and an HSV colour-pattern filter, per-camera area thresholds); and (b) WASB/HRNet [4], a pretrained heatmap-based detector with a basketball-specific checkpoint, used zero-shot with a confidence threshold recalibrated empirically (well below the broadcast-tuned default). On the widest-shot camera (`out13`, ball ≈ 7 px after resize) WASB is additionally run with 3×3 overlapping tiling.

2.4 Cross-Camera Association and 3D Triangulation

Per-camera tracks are undistorted using the calibration from Sec. 2.2. For every camera pair and frame, candidate matches between per-camera track IDs are scored by triangulating each detection pair and measuring the combined reprojection error in both views [1]; a Hungarian assignment selects the best pairing per frame (reprojection error < 30 px). A track-level global identity forms only from pairs that are *mutually* the best match and win the per-frame assignment in ≥ 5 distinct frames; consistent pairs merge into cross-camera groups via union-find, one global ID per group.

For every frame, detections sharing a global ID from ≥ 2 cameras are triangulated with a multi-view DLT solve. A court-sized plausibility box (28×15 m $+5$ m margin, height $[-1, 3]$ m) discards points outside it, removing residual wrong matches rather than genuine players.

3 Results

3.1 Experimental Setup

The 2D evaluation matches predicted/ground-truth boxes per frame (per camera, per class) with a Hungarian assignment on IoU, reporting Precision, Recall, F1 and MOTP (mean IoU over matched pairs), following CLEAR MOT [5]. The acceptance threshold is deliberately low (any non-zero overlap counts as a candidate match), so Precision/Recall/F1 mainly reflect whether an object was found in roughly the right place, while MOTP carries the localization-quality signal. The 3D evaluation triangulates annotated boxes (≥ 2 views, same frame, same category) into a pseudo-ground-truth 3D point, matched to the reconstruction with a Hungarian assignment on 3D distance (threshold 3 m).

3.2 2D Tracking

Table 1 reports player/referee accuracy per camera and the ball trackers pooled over all three (per-camera ball numbers follow the same `out13`-is-hardest pattern). `out2`/`out4` (closer, narrower shots) track players considerably better than `out13` (wide, more/smaller/more-occluded people); pooled, Recall = 0.712, Precision = 0.775, MOTP = 0.423. For the ball, WASB roughly doubles F1 over the classical baseline with fewer false positives, but both miss most instances – even WASB finds fewer than 1 in 6 – consistent with the small-object argument above.

Table 1: 2D tracking accuracy: players+referees pooled (class_id 0) per camera and aggregated; ball trackers (class_id 1) pooled over all three cameras.

		TP	FP	FN	Precision / Recall / F1	MOTP
Players	`out2`	583	44	179	0.930 / 0.765 / 0.839	0.441
	`out4`	459	98	172	0.824 / 0.727 / 0.773	0.470
	`out13`	848	408	414	0.675 / 0.672 / 0.674	0.386
	All	1890	550	765	**0.775 / 0.712 / 0.742**	**0.423**
Ball	Classic (MOG2+HSV)	16	263	175	0.057 / 0.084 / 0.068	0.116
	WASB/HRNet	25	133	166	0.158 / 0.131 / 0.143	0.083

3.3 3D Reconstruction

Of 1085 pseudo-ground-truth 3D points, 487 (44.9% coverage) match a reconstructed point within 3 m, with an overall MED of 0.722 m and RMSE of 1.013 m. Per-identity (Table 2), most players/referees localize with sub-meter accuracy, with two exceptions: the ball (only 3 matches, limited by the same detection scarcity as Sec. 3.2) and `Red_23` (1.91 m over 75 matches – not a handful of outliers, pointing to a systematic cross-camera association problem for that identity).

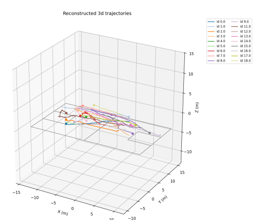

Fig. 1: Reconstructed 3D trajectories, court-centered frame; each colour is one global cross-camera identity (Sec. 2.4).

Table 2: 3D error by identity, sorted by mean error (two columns for space).

Identity	Mean [m]	n	Identity	Mean [m]	n
`Red_2`	0.307	29	`White_14`	0.592	26
`Refree_1`	0.308	49	`White_2`	0.614	32
`Red_9`	0.381	14	`Red_7`	0.668	11
`White_27`	0.460	45	`White_34`	0.678	71
`White_16`	0.476	82	`Ball`	1.431	3
`Red_11`	0.496	50	`Red_23`	1.907	75

3.4 Trajectory Continuity

Point accuracy says nothing about identity continuity over time, which task 3a asks for explicitly. For each of the 18 global 3D tracks, *continuity* is the fraction of frames within its own first-to-last-frame span with a reconstructed position

(1.0 = no gaps): 88.1% on average (median span 155 frames = 6.2 s; mean 3.4 gaps/track). More tellingly, 18 IDs exceeds the ∼12–14 people on court (10 players + 2 referees): vote-based re-identification (Sec. 2.4) sometimes fragments one person into several global tracks once cross-camera correspondence is lost, and never re-merges them – a limitation of an association scheme with no appearance model.

4 Conclusions and Future Work

We built and evaluated a three-camera pipeline for player/referee and ball tracking with 3D reconstruction, covering the mandatory tasks for a two-person team using all three views rather than the two required. Players/referees reach moderate-to-good 2D accuracy (pooled F1 = 0.742) and sub-meter 3D median error (0.72 m), validating the pipeline for the common case; the ball and `Red_23` are the two clear outliers. The ball proved hard throughout: a fine-tuned-detector-plus- tiling attempt was abandoned once diagnosed as structurally limited by object size, and even the better pretrained model (WASB) stays well short of usable.

Future work: diagnose the `Red_23` outlier (likely a recurring wrong cross-camera pairing); add appearance-based re-identification against the fragmentation of Sec. 3.4; revisit ball detection at its true pixel scale; adopt identity-aware MOT metrics (MOTA, ID switches); and, as a bonus, import the motion into Unreal Engine (task 4).

References

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